



ACO-IM: maximizing influence in social networks using ant colony optimization

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Abstract

Online social networks play an essential role in propagating information, innovation, and ideas via word-of-mouth spreading. This word-of-mouth phenomenon leads to a fundamental problem, known as influence maximization (IM) or subset selection problem. The IM problem aims to identify a small subset of users, viz. seed nodes such that overall influence spread can be maximized. The seed selection problem is NP-hard, unfortunately. A greedy solution of IM problem is not sufficient due to the use of time-consuming Monte Carlo simulations, which is limited to small-scale networks. However, the greedy solution ensures a good approximation guarantee. In this paper, a local influence evaluation heuristic is adopted to approximate local influence within the two-hop area. With this heuristic, an expected diffusion value under the traditional diffusion models is evaluated. To optimize local influence evaluation heuristic, an influence maximization algorithm based on ant colony optimization (ACO-IM) is presented. ACO-IM redefines the representation and updates the rule of pheromone deposited by ants and heuristic information. The algorithm uses the probabilistic environment to avoid premature convergence. Finally, the experimental results show the superiority of the proposed algorithm. The statistical tests are also performed to distinguish the proposed method from the state-of-the-art methods.

Keywords Information diffusion · Influence maximization · Social networks · Ant colony optimization

1 Introduction

Online social networks such as Twitter, Facebook, LinkedIn, email networks, scientific collaboration networks, and microblogs appear as a social structure to connect individuals or organizations with the *word-of-mouth* effect (Brown and Reingen 1987; Goldenberg et al. 2001). This word-of-mouth spreads influence or information upon the social network. Information diffusion is a fundamental issue to analyze and

investigate the social network. The influence maximization (IM) problem is constructed based on this issue. IM is the problem of addressing a small subset of influential users that can maximize the influence spread in the network. The influential user's selection problem has many applications like viral marketing (Domingos and Richardson 2001), revenue maximization (Teng et al. 2018), social recommendation (Ye et al. 2012), rumor control (Wu and Pan 2017; Chen et al. 2010b), network monitoring (Leskovec 2007), etc.

Let us recall the example of the IM problem presented by Kempe et al. (2003) in the field of viral marketing. Consider a social network as a platform for influence propagation with the individual's extent of influence to another. An advertising company propagates the information of a new product in the network with the aims to maximize the adoption of the product. Initially, the company targets a small subset of individuals (seed users) by providing free samples of the product (due to the limited budget of marketing). The marketer aims those seed users to recommend the product to their friends via word-of-mouth strategy so that maximum adoption of the product can be possible. Therefore, this viral marketing strategy introduces the IM problem based on word-of-mouth

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propagation model. Inspired by the above scenario, Domingos and Richardson (2001) were first to introduce influence maximization (IM) as an optimization problem. Later, IM problem is formally formatted by Kempe et al. (2003), given as $S^* = \operatorname{argmax}_{|S|=k} \{\sigma(S)\}$, where $\sigma(S)$ is an objective function which estimates the expected adoption of the product in the network. IM problem is also known as the k -node subset selection problem. Kempe et al. proved that the objective function $\sigma(S)$ is sub-modular. They also prove that IM is *NP-hard* under traditional diffusion models. The greedy solution presented by Kempe et al. provides approximation guarantee within a factor of $(1 - 1/e - \epsilon)$.

All the existing IM algorithms have their own weaknesses. The greedy approaches (Leskovec et al. 2007; Wang and Feng 2009; Wang et al. 2010; Goyal et al. 2011) provide a good-quality seed. To estimate accurate marginal influence spread, these approaches rely on time-consuming Monte Carlo (MC) simulations. Therefore, the efficiency of greedy approaches limits its applicability on large-scale network. In order to improve scalability, heuristic approaches (Kimura and Saito 2006; Chen et al. 2009, 2010a, b) are introduced. Heuristic approaches scarify some degree of accuracy to gain high efficiency and scalability. To maintain a trade-off between accuracy and convergence, mixed approaches (Chen et al. 2010a; Jung et al. 2012; Kim et al. 2013) are introduced. These approaches are just a compromise between efficiency and accuracy.

This study borrowed ant colony optimization (ACO) (Dorigo and Stützle 2003; Dorigo and Blum 2005) from nature-inspired swarm optimization techniques to solve the IM problem. We have some advantages of using ACO over other swarm optimization techniques like inherent parallelism, easy to adopt dynamics (degree, distance, topology), contextual (spatial, temporal, competitive) features, etc. ACO is combining the power of general meta-heuristics with problem-specific heuristics to find an approximate solution within a reasonable computing time. The solution efficiency and search speed of basic ACO algorithm lead to premature convergence. Therefore, to avoid premature convergence, we use an improved ACO to solve our combinatorial problem. The motivation behind the use of a nature-inspired optimization to solve the IM problem is given as follows:

1. There are several algorithms present in the literature to solve combinatorial optimization problems. However, most of these algorithms are following the instance-based search, i.e., future solutions evolved using the current instance of solution. In contrast, nature-inspired meta-heuristic algorithms are model-based search, i.e., future solutions evolved using a probabilistic model which is based on the current instance of the solution (Zlochin et al. 2004).

2. Some diffusion models like the competitive threshold model may not be sub-modular and monotonic, which leads to the inapplicability of greedy-based methods. Therefore, we need to address the IM problem for a wide range of diffusion models.
3. It is suitable for those problems who have intractable NP-hard issues due to its simplicity and efficiency (Dorigo and Di Caro 1999).

Contribution In this paper, we introduce an influence maximization algorithm based on ant colony optimization (ACO-IM) for seed selection. The major contributions of our work are as follows.

- We extend the local influence heuristic *expected diffusion value* (EDV) (Jiang et al. 2011) to approximate local influence within the two-hop area as $EDV_{[2]}(S)$.
- We present an ACO approach to optimize local influence evaluation function $EDV_{[2]}(S)$ by redefining the representation and update rules for pheromone and heuristic information. A degree-based solution initialization method is also presented.
- The theoretical, as well as experimental, comparison of ACO-IM is presented against the state-of-the-art methods. The experimental study analyzes the performance of the proposed algorithm in terms of quality and efficiency on various real-world social networks under traditional diffusion models. The statistical tests are also performed to demonstrate the significant difference between ACO-IM and state-of-the-art methods.
- The experimental results illustrate that the proposed algorithm outperforms the state-of-the-art techniques except for CELF++ and LAPSO-IM in terms of influence spread. Similarly, ACO-IM is more efficient than CELF++ and comparable with remaining algorithms.

Organization Section 2 illustrates the related study in the field of IM problem along with recent developments. Section 3 explains the preliminary information and defines the problem statement. Section 4 presents the proposed work, and Sect. 5 discusses the working of the proposed algorithm along with time complexity analysis. Section 6 presents the experimental analysis of the proposed algorithm against the state-of-the-art methods on six real-world social networks. Finally, Sect. 7 discusses the conclusion of our findings with directions for future work.

2 Related work

Domingos and Richardson (2001) were the first to introduce IM as an algorithmic problem and develop a probabilistic model to solve the IM problem. Later, Kempe et al.

(2003) format the IM problem as an optimization problem and present a greedy algorithm to find seed users for maximizing the influence spread in the network. All the existing IM algorithms can be divided into three classes based on their performance. Greedy-based, heuristic-based, and mixed approaches aim to provide effective, efficient, and a trade-off between both, respectively.

2.1 Greedy-based approaches

The greedy-based approaches (Leskovec et al. 2007; Wang and Feng 2009; Wang et al. 2010; Goyal et al. 2011) provide a good-quality seed. To estimate accurate marginal influence spread, these approaches rely on time-consuming MC simulations. Therefore, the efficiency of greedy approaches limits its applicability on a large-scale network. Kempe et al. (2003) are very first to propose a greedy solution for IM problem. They state that the IM problem is NP-hard under traditional diffusion models and also provides an approximate guarantee of at least 63% of the optimal solution. Any diffusion models can adopt the greedy algorithm. Later, Sviridenko (2004) extends greedy framework presented by Kempe et al. (2003) using non-uniform cost function for seed selection. The behavior of objective function $\sigma(S)$ is still sub-modular. Inspired by the idea of greedy framework, Leskovec et al. (2007) proposed a novel algorithm, viz. *cost-effective lazy forward* (CELFF), by incorporating the sub-modular property of the objective function. CELF significantly reduces the search space of the greedy algorithm by storing marginal gain of every individual x with current seed set S . Therefore, CELF is almost 700 times efficient than greedy with approximate influence spread. Goyal et al. (2011) improvise the CELF algorithm and present an extended method named as CELF++. They further reduce the search space of CELF by storing marginal gain of every individual x with current seed set S along with $\{y\} \cup S$, where y is the highest marginal gain node till now. Therefore, CELF++ is 30–50% efficient than CELF with approximate influence spread. Wang and Feng (2009) proposed a new algorithm named as *targeted wise greedy* (TW) based on node potential.

2.2 Heuristic-based approaches

In order to improve scalability, heuristic-based (Kimura and Saito 2006; Chen et al. 2009, 2010a,b) and meta-heuristic (Gong et al. 2016; Singh et al. 2019a,b) IM approaches are introduced. Heuristic-based approaches scarify some degree of accuracy to gain high efficiency and scalability and have the benefit of practical efficiency. MaxDegree (Kempe et al. 2003) is a straightforward and efficient method to select the most influential users based on their node degree. This method generates a low-quality seed set because it only considers topological information and avoids influence overlap,

path, community information, etc. Inspired by MaxDegree, Chen et al. (2009) introduced a method called *degree discount* (DD) by considering degree discount step. Degree discount step states that when a node x activates its neighbors y , y is no longer able to influence or re-activate node x . Therefore, DD step reduces the degree of node y by one, although this improvement is minimal. Some other heuristics based on influence path like SP1M (Kimura and Saito 2006), SIM-PATH (Goyal et al. 2011), LDAG (Chen et al. 2010b), and IPA (Kim et al. 2013) have been proposed. Kundu et al. (2011) proposed a centrality measure known as diffusion degree based on local influence evaluation function. The local influence evaluation function is defined as the sum of the node's direct influence on their neighbors and indirect influence on the rest of the network via neighbors. Other heuristic-based approaches include CGA (Wang et al. 2010), CIM (Chen et al. 2014), CoIM (Singh et al. 2019c), etc.

2.3 Mixed approaches

To maintain a trade-off between accuracy and convergence, mixed approaches (Chen et al. 2010a; Jung et al. 2012; Kim et al. 2013) are introduced. These approaches are just a compromise between efficiency and accuracy. With the idea of SP1M, Chen et al. (2010a) presented a new influence path-based approach known as *maximum influence arborescence* (MIA). This method iteratively selects the maximal propagation probability path based on local arborescence structure. MIA is much efficient than SP1M. Jung et al. (2012) were first to introduce robustness in IM solution and presented a scoring-based method called IRIE. This method works in two phases: influence ranking (IR) and influence estimation (IE). The IR phase estimates the ranking of each node based on the message passing mechanism. Then, IE phase evaluates the influence probability of each node. Finally, IRIE iteratively selects seed nodes based on IR/IE updation. Other mixed approaches include PRUNEDMC (Ohsaka et al. 2014), TIM (Tang et al. 2014), StaticGreedy (Cheng et al. 2013), etc.

Recently, context-aware IM techniques have been presented to improve the effectiveness of seed nodes. Some research work has been proposed by extending the classical IM problem with the addition of contextual features, such as location (Guo et al. 2017; Li et al. 2014), topic (Li et al. 2015; Chen et al. 2015; Singh et al. 2019d), competitive (Zhu et al. 2016; Lin et al. 2015), and time (Gomez-Rodriguez et al. 2016; Gomez Rodriguez and Schölkopf 2012). Some symmetric and *nonnegative latent factor* (NLF) models (Hu et al. 2019; Luo et al. 2019; Hu et al. 2018; Luo et al. 2016, 2017) are introduced to extract useful information from networks and applied in industrial applications. Those models can be helpful to develop high computational efficiency methods in viral marketing and recommender system. Tables 1 and 2 present the comparative analysis of the pro-

Table 1 Theoretical comparative analysis of influence maximization algorithms against the proposed method (Singh et al. 2019b)

Algorithm	Time complexity	Approximation	Problem solving Perspective	State-of-the-art algorithms	Base algorithm
Greedy (Kempe et al. 2003)	$O(kNMI)$	$1 - 1/e - \epsilon$	Spread simulation	MaxDegree, Central & Random	—
Knapsack Greedy (Sviridenko 2004)	$O(N^5)$	$1 - 1/e - \epsilon$	Spread simulation	—	Greedy
SPIM (Kimura and Saito 2006)	$O(kNM)$	$1 - 1/e$	Influence path	Degree, PageRank & Closeness	—
CELF (Leskovec et al. 2007)	$O(kNMI)$	$1 - 1/e - \epsilon$	Sub-modularity	Greedy	Greedy
Degree discount (Chen et al. 2009)	$O(k \log N + M)$	N.A.	Spread simulation	CELF, Greedy & Random	High Degree
NewGreedy (Chen et al. 2009)	$O(kIM)$	N.A.	Snapshots	CELF, Greedy & Random	High Degree
TW Greedy (Wang and Feng 2009)	$O(kNMI)$	$1 - 1/e - \epsilon$	Spread simulation	SCG, KKG & High Degree	Greedy
MIA/PMIA (Chen et al. 2010a)	$O(Nt_{i\theta} + kn_{\theta}n_{i\theta}(n_{i\theta} + \log N))$	$1 - 1/e$	Influence path	Greedy, Random, DD & PageRank	SPIM
LDAG (Chen et al. 2010b)	$O(Nt_{i\theta} + kn_{\theta}m_{i\theta}(m_{i\theta} + \log N))$	N.A.	Score estimation	Greedy, SPIN, DD & PageRank	—
CGA (Wang et al. 2010)	$O(M + IM_C(N(Z - C) + k(C + N_C)))$	$1 - e^{-1/(1+\delta_C)}$	Community based	DD, MG & Random	—
CELF++ (Goyal et al. 2011)	$O(kNMI)$	$1 - 1/e - \epsilon$	Sub-modularity	CELF	CELF
Diffusion degree (Kundu et al. 2011)	$O(N + M)$	N.A.	Centrality based	DD & High Degree	High Degree
SIMPATH (Goyal et al. 2011)	$O(kINP_{\theta})$	N.A.	Score estimation	High Degree, CELF & PageRank	LDAG
IRIE (Jung et al. 2012)	$O(k(n_{\theta}k + M))$	N.A.	Score estimation	Greedy & PMIA	—
IPA (Kim et al. 2013)	$O(\frac{NO_n n_{uu}}{c} + k^2(\frac{O_n n_{uu}}{c} + (c - 1)))$	N.A.	Influence path	Greedy, DD & Random	PMIA
StaticGreedy (Cheng et al. 2013)	$O(\frac{kMN^2 \log(\frac{N}{k})}{\epsilon^2})$	$1 - 1/e - \epsilon$	Snapshots	CELF, SPIM, DD & High Degree	PMIA
PRUNEDMC (Ohsaka et al. 2014)	$O(\frac{kMN^2 \log(\frac{N}{k})}{\epsilon^2})$	$1 - 1/e - \epsilon$	Snapshots	IRIE, Random, PMIA & Degree	Greedy
TIM (Tang et al. 2014)	$O(\frac{k(M+N) \log N}{\epsilon^2})$	$1 - 1/e - \epsilon$	Reverse Reachability	CELF++, IRIE & SIMPATH	—
DPSO (Gong et al. 2016)	$O(k^2 \log kn \bar{D}^2)$	N.A.	Swam optimization	Degree, CELF++ & SAEDV	PSO
LAIM (Ge et al. 2017)	—	$1 - 1/e - \epsilon$	Learning Based	Degree, CELF & Random	Greedy
MIM2 (Singh et al. 2019e)	$O((l + m)(N + M) + (k + m)(N \log N + M))$	N.A.	Rank Refinement	DD, MaxDegree, CELF & Random	MIM-Greedy
LAPSO-IM (Singh et al. 2019b)	$O(lnk(\log k + \bar{D}^2) + nkN)$	N.A.	Swam optimization	DD, CELF++ & DPSO	PSO
IM-SSO (Singh et al. 2019a)	$O(n(kD_{\text{avg}}(N)^2 + n) + n \log n)$	N.A.	Swam optimization	DD, CELF++ & PSO	SSO
ACO-IM	$O(I_{\max} R_G V + V \log V)$	N.A.	Swam optimization	DD, CELF++, IM-SSO & PSO	ACO

Table 2 Comparison of the characteristics of the baseline methods against the proposed algorithm (Singh et al. 2019b)

Algorithm	Diffusion model				Category			Network		
	Linear threshold	Independent cascade	Triggering	Continuous time-aware	Simulation	Heuristic	Meta-heuristic	Sketch	Single	Static
Greedy (Kempe et al. 2003)	✓	✓	✓	✓	✓	×	×	×	✓	✓
Knapsack Greedy (Sviridenko 2004)	✓	✓	✓	✓	✓	×	×	×	✓	✓
SPIM (Kimura and Saito 2006)	×	✓	×	×	×	✓	×	×	✓	✓
CELF (Leskovec et al. 2007)	✓	✓	✓	✓	✓	×	×	×	✓	✓
Degree discount (Chen et al. 2009)	✓	✓	✓	✓	×	✓	×	×	✓	✓
NewGreedy (Chen et al. 2009)	×	✓	×	×	×	×	×	✓	✓	✓
TW Greedy (Wang and Feng 2009)	✓	✓	✓	✓	✓	×	×	×	✓	✓
MIA/PMIA (Chen et al. 2010a)	×	✓	×	×	×	✓	×	×	✓	✓
LDAG (Chen et al. 2010b)	✓	×	×	×	×	✓	×	×	✓	✓
CGA (Wang et al. 2010)	×	✓	×	×	✓	×	×	×	✓	✓
CELF++ (Goyal et al. 2011)	✓	✓	✓	✓	✓	×	×	×	✓	✓
Diffusion degree (Kundu et al. 2011)	✓	✓	✓	✓	×	✓	×	×	✓	✓
SIMPACT (Goyal et al. 2011)	✓	×	×	×	×	✓	×	×	✓	✓
IRIE (Jung et al. 2012)	×	✓	×	×	×	✓	×	×	✓	✓
IPA (Kim et al. 2013)	×	✓	×	×	×	✓	×	×	✓	✓
StaticGreedy (Cheng et al. 2013)	×	✓	×	×	×	×	×	✓	✓	✓
PRUNEDMC (Ohsaka et al. 2014)	×	✓	×	×	×	×	×	✓	✓	✓
TIM (Tang et al. 2014)	✓	✓	✓	×	×	×	×	✓	✓	✓
DPSO (Gong et al. 2016)	✓	✓	✓	✓	×	×	✓	×	✓	✓
LAIM (Ge et al. 2017)	✓	✓	✓	✓	✓	×	×	×	✓	✓
MIM2 (Singh et al. 2019e)	✓	✓	✓	✓	×	✓	×	×	×	✓
LAPSO-IM (Singh et al. 2019b)	✓	✓	✓	✓	×	×	✓	×	✓	×
IM-SSO (Singh et al. 2019a)	✓	✓	✓	✓	×	×	✓	×	✓	×
ACO-IM	✓	✓	✓	✓	×	×	✓	×	✓	✓

Table 3 Notations

$G(V, E)$	$\triangleq$	A social network with vertex set V and edge set E
$N(u)$	$\triangleq$	The neighbors set of node u
S	$\triangleq$	Seed set
k	$\triangleq$	The number of nodes in seed set ($ S $)
$\sigma(S)$	$\triangleq$	The expected influence of seed set in the network, i.e., $Inf(S)$
pop	$\triangleq$	The population of ants in the colony
$Solution(A_i)$	$\triangleq$	The solution vector of the ant A_i
$I_{\max}$	$\triangleq$	The maximum number of iterations
$\tau_I(u)$	$\triangleq$	The pheromone value of node u at iteration I
ρ	$\triangleq$	The evaporation rate parameter
$\eta(u)$	$\triangleq$	The quality value of the node u
$f(Solution(A_i))$	$\triangleq$	The fitness value of the solution generated by A_i

posed method against the baseline methods. Table 1 discusses the theoretical analysis of IM approaches and provides various parameters like time complexity, approximation ratio (N.A. indicates no approximation guaranteed), base algorithms, state-of-the-art methods, etc. Table 2 illustrates the characteristics of the IM approaches like adaptive diffusion models ($\checkmark$ = work and $\times$ = not work), algorithmic category ($\checkmark$ = belongs and $\times$ = not belongs), networks ($\checkmark$ = applied and $\times$ = not applied).

3 Preliminaries

3.1 Notations

Notations that are needed for problem formulations in this paper are given in Table 3.

3.2 Definitions

Definition 1 (Social Network) A social network with N users and M social ties is represented as a weighted-directed graph $G(V, E, W)$. Here, V denotes a set of users, $|V| = N$, and E represents a set of relationships, $|E| = M$, and W represents the edge weight. A social network is also known as an influence graph.

Definition 2 (Neighbors) Neighbors $N(u)$ of node u are defined as the set of users v such that $v \in N(u)$ iff $\exists(u, v) \in E$, $v \in V$. $N_{\text{inc}}(u)$ and $N_{\text{out}}(u)$ denote in and out neighbors of node u , respectively.

Definition 3 (Degree centrality) Degree centrality is defined as the number of links incident upon a node, i.e., $C_D(u) = |N(u)|$. In ACO-IM problem, we consider node degree as $C_D(u) = |N_{\text{out}}(u)|$.

Definition 4 (Seed nodes) Seed nodes (S) are the set of nodes who act as the source of the information propagation process in the social network, $|S| = k$, $S \in V$.

Definition 5 (Active node) A node $u \in V$ is called active if either $u \in S$ or u adopted the information propagated by previously active nodes $v \in V_A$ under diffusion model M . Once u is activated, then $V_A \leftarrow \{V_A \cup u\}$.

Definition 6 (Influence spread) Influence spread $I_S(S)$ of the seed set S is defined as the number of active users after diffusion process under a diffusion model, i.e., $I_S(S) = |V_A(S)|$.

3.3 Diffusion model

Many information diffusion models are studied and presented in the field of social network analysis. In this paper, the three most commonly used tradition models are utilized for information spreading: *independent cascade* (IC), *weighted cascade* (WC), and *linear threshold* (LT). In each of the diffusion models, every node belongs to one of the two states but not both, i.e., node either belongs to an active or inactive state. Every node is inactive except seed nodes at the start of the diffusion process. Only active nodes can influence or activate their inactive neighbors. Once a node becomes active, it will never change its state. The diffusion process continues until no new inactive nodes become active.

In the independent cascade model, every active node has only one chance to influence its inactive neighbors. For example, a node x becomes active at time t and then tries to influence its inactive neighbors y with influence probability $p_{x,y}$ at next discrete time, i.e., $(t + 1)$. The weighted cascade model is the special case of IC model presented by Kempe et al. (2003). The main difference between IC and WC model lies in the way of determining influence probability of an edge. In general, the IC model follows a constant ($p_{x,y} \in [0.01, 0.1]$) or trivalency model ($p_{x,y} \in \{0.001, 0.01, 0.1\}$) for assigning influence probability for edge (x, y) . The WC

model determines the influence probability based on indegree of y , i.e., $p_{x,y} \in \frac{1}{\text{indegree}(y)}$. In the linear threshold model, each node x has a threshold θ_x as the activation threshold. A node y becomes active iff the sum of influence probabilities of active incoming neighbors, i.e., $\sum_{x \in N_{\text{inc}}^A(y)} p_{x,y} \geq \theta_y$. $N_{\text{inc}}^A(y)$ denotes the active incoming neighbors of y . The influence probability in this model can be assigned as WC model.

3.4 Ant colony optimization

The *ant colony optimization* (ACO) is an optimization approach to find an approximate solution in a reasonable amount of time for combinatorial optimization problem (Dorigo and Stützle 2003; Dorigo and Blum 2005). The ACO algorithms are inspired by the real ants and their foraging behavior. The core of the communication strategy among ants is the cooperative behavior of ants using chemical pheromone trails. In ACO approaches, a finite set of artificial ants explore the surrounding area of their nest and search for good-quality solutions under consideration of an optimization problem. Initially, each ant builds its own solution based on problem-dependent criteria and evaluates the quality of the solution. The quantity of the pheromone value deposited by an ant helps the other ants to measure the quality of the solution. The ants use an incremental constructive approach to find a feasible solution. The three major processes of the ACO algorithm are: construct ant solutions, update pheromones, and daemon actions.

3.5 Problem definition

Given an influence graph $G = (V, E)$, a diffusion model, and an integer k , the influence maximization process selects a seed set $S \in V$ to maximize the influence spread in G , i.e.,

$$\sigma(S) = \operatorname{argmax}_{S^* \subseteq V \wedge |S^*| = k} \sigma(S^*). \quad (1)$$

4 Proposed work

In this section, we present a single objective algorithm ACO-IM for the objective of maximizing the influence in the social network. The proposed algorithm ACO-IM conducts to explore the network and finds the set of most influential users S , $|S| = k$. ACO-IM is divided into three procedures given as follows:

1. **Construct ant solutions** This procedure manages a colony of ants to explore the nest for finding feasible solutions. These ants apply stochastic search process to find solution using pheromone trails and quality information. The solution of each ant is constructed iteratively in the same way.

2. **Pheromones update** This procedure updates the value of the pheromone trails. The value of pheromone trails increases when ants deposited the pheromone and decreases due to evaporation.
3. **Daemon actions** This procedure performs local or global optimization to approximate the solution. This local optimization function decides whether there is a need of additional pheromone or not.

The principal components of the ACO-IM are outlined as follows.

- **Ants** The ants are the agents of the algorithm to perform optimization. ACO-IM initializes the population of spiders in the web at the beginning of the algorithm. Here, the population is assigned with $|D(u)| \geq D_{\text{Avg}}(G)$, $\forall u \in V$.
- **Fitness assignation** In order to estimate the accurate influence spread of seed S in social network efficiently, Jiang et al. (2011) proposed a local influence function *expected diffusion value* (EDV) under IC model. EDV considers direct influence of a node u to its neighbors. The expected diffusion value of seed set S in the one-hop area is computed as follows:

$$\text{EDV}(S) = \sigma_0(S) + \sigma_1(S) = k + \sum_{u \in S} (1 - \prod_{(u,v) \in E, v \in N^{(1)}(S) \setminus S} (1 - p_{uv})), \quad (2)$$

where p_{uv} represents the influence probability of u to v . In social networks, the influence spread of a user opinion is limited to within its local region like three-hop area (Christakis and Fowler 2009) and two-hop area (Pei et al. 2014). Following these studies, we present an approximate influence estimation function $\text{EDV}_{[2]}(S)$ within the two-hop area. We can estimate the expected diffusion value in two-hop area $\text{EDV}_{[2]}(S)$ as follows:

$$\begin{aligned} \text{EDV}_{[2]}(S) &= \sum_{i=0}^{i=2} \sigma_i(S) = k + \sigma_1(S) \\ &+ \sum_{v \in N^{(1)}(S) \setminus S} (1 - \prod_{(v,w) \in E, w \in N^{(2)}(S) \setminus \{N^{(1)}(S)\}} p_A(v) \cdot (1 - p_{vw})) \end{aligned} \quad (3)$$

$$\begin{aligned} \text{EDV}_{[2]}(S) &= k + \sigma_1(S) \\ &+ \frac{\sigma_1(S)}{|N^{(1)}(S) \setminus S|} \sum_{w \in N^{(2)}(S) \setminus S} p_A(w) \cdot D_{1-2}(w) \end{aligned} \quad (4)$$

$$\begin{aligned} \text{EDV}_{[2]}(S) &= k \\ &+ \left(1 + \frac{1}{|N^{(1)}(S) \setminus S|} \sum_{w \in N^{(2)}(S) \setminus S} p_A(w) \cdot D_{1-2}(w) \right) \end{aligned}$$

$$\left(\sum_{u \in S} \left(1 - \prod_{(u,v) \in E, v \in N^{(1)}(S) \setminus S} (1 - p_{uv}) \right) \right), \quad (5)$$

where $N^1(S)$ and $N^2(S)$ denote the one-hop area friends and the two-hop area (friends of friends) of S , respectively. $p_A(u)$ is the active probability of node u . $D_{1-2}(w)$ represents the number of connections within $N^1(S)$ and $N^2(S)$. The fitness function of ACO-IM is given by local influence spread function $EDV_{[2]}(S)$ depicted in Eq. 5.

- **Pheromone value** The pheromone value describes the solution quality. Initially, the pheromone value of nodes is initialized with the local influence of top $p\%$ solution. Let $\tau(u)$ and $\rho \in (0, 1]$ are the pheromone value on node u and evaporation rate of the pheromone, respectively. If I is the iteration number, then pheromone $\tau(u)$ of node u can be updated as follows:

$$\tau_{I+1}(u) \leftarrow (1 - \rho) \cdot \tau_I(u) + \frac{\rho}{I} \cdot \sum_{l=1}^{|S_{\text{best}}|} \tau(u)^l, \quad (6)$$

where

$$\tau(u)^l \leftarrow \begin{cases} f(S_l) & \text{if } u \in S_l, S_l \in S_{\text{best}} \\ 0 & \text{Otherwise} \end{cases}. \quad (7)$$

- **Heuristic information** The heuristic value is estimated based on fitness function, i.e., local influence in ACO-IM. Let $\tau(u)$ and ρ are the pheromone value on node u and evaporation rate of the pheromone, respectively. The quality $\eta(u)$ of node u is evaluated as follows:

$$\eta(u) \leftarrow \rho \cdot \tau(u). \quad (8)$$

- **Solution** Each ant A_i maintains a vector $Solution(A_i) = (s_1, s_2, \dots, s_k)$, $|Solution(A_i)| = k$. The solution vector is updated in each iteration using Eq. 9. Let α and β are relative importance of the pheromone and the heuristic information or quality of node. An ant A_i positioned at node x decides to move at node y by applying the rule given as follows:

$$y \leftarrow \begin{cases} \text{argmax}_{v \in V \setminus x} \{ [\tau(v)]^\alpha \cdot [\eta(v)]^\beta \} & \text{if } r < r_0 \\ N & \text{Otherwise} \end{cases} \quad (9)$$

where $N \in V$ is a random node selected based on the probability distribution given as follows:

$$p(u) \leftarrow \frac{[\tau(u)]^\alpha \cdot [\eta(u)]^\beta}{\sum_{w \in V} [\tau(w)]^\alpha \cdot [\eta(w)]^\beta}. \quad (10)$$

4.1 Premature convergence avoidance

ACO-IM may stuck in local optima, i.e., there is a need to avoid premature convergence. Therefore, we incorporate the probabilistic parameter $p_I \in (0, 1)$ which is defined as follows:

$$p_I = \frac{\log(I)}{\log(I_{\max})}, \quad (11)$$

where I and $I_{\max}$ represent the current iteration number and the maximum number of iterations, respectively. In each iteration, p_I is compared with a randomly generated value $p_{\text{rand}} \in [0, 1]$. The possibility of event $p_{\text{rand}} > p_I$ decreases with an increase in the value I because it increases the value of p_I . Similarly, the possibility of event $p_{\text{rand}} > p_I$ increases with the decreasing value of I . Therefore, the possibility of searching new node is high in initial iterations and vice versa for later iterations. Hence, ACO-IM prevents the premature convergence using event $p_{\text{rand}} > p_I$.

4.2 Stagnation avoidance

ACO-IM may suffer from another undesirable situation, i.e., stagnation. The stagnation situation arises when almost every ant A_i generates the same solution repeatedly. This happens because of the inappropriate tuning of parameters ρ , α , and β . The probability of stagnation increases with the increase in ρ parameter value. Therefore, to avoid this situation we set the parameter values $\rho = 0.2$, $\alpha = 0.5$, and $\beta = 0.5$.

4.3 Convergence test

We need to perform the convergence test for addressing the problem if ACO-IM finds the optimal solution when enough resources are available. The pheromone update rule may prevent the algorithm to achieve optimal solution due to stochastic nature of search procedure. There is a need to consider two types of convergence of ACO-IM: *convergence in solution* (reachability convergence) and *convergence in value* (asymptotic convergence) (Dorigo and Blum 2005). Reachability convergence estimates the probability of an algorithm to repeatedly generate the same optimal solution. Asymptotic convergence estimates the probability of an algorithm to find optimal solution at least once.

Proposition 1 If $\tau_I(u)$ and $I_{\max}$ are the pheromone value of node u at iteration I and maximum number of iterations, respectively, then using Eq. 6 pheromone update rule ACO-IM holds the following:

$$\lim_{I \rightarrow \infty} \{ \tau_I(u) \} \leq \frac{\tau(u) \cdot I_{\max}}{\rho}. \quad (12)$$

Algorithm 1: ACO-IM(G,k): Ant colony optimization-based influence maximization algorithm

Input: Influence graph G , seed size k .
Output: Seed set S .

- 1 $I \leftarrow 0$ ▷ Initialization phase
- 2 $\forall u \in V, \tau_0(u) \leftarrow \epsilon$ ▷ Pheromone value of node
- 3 $S_{best} \leftarrow \phi$ ▷ Set of top $p\%$ solutions
- 4 $Solution(A_i) \leftarrow \phi; |Solution(A_i)| = k; \forall A_i \in A$ ▷ Solution of ant A_i
- 5 Distribution of ants over the influence graph ▷ Initialize the population
 - $D_{Avg}(G) \leftarrow$ Find average degree of the network
 - $R_G \leftarrow$ Identify each node $u \in V$ with degree $D(u) \geq D_{Avg}(G)$
 - Distribute ants on nodes $u \in R_G$
- 6 Initialize the solution for each ant $A_i \in A, |A| = |R_G|$ ▷ Initialize the solution
- for each ant** $A_i \in A$ **at node** $u \in R_G$ **do**
 - $Solution(A_i) \leftarrow u$
 - $Solution(A_i) \leftarrow Solution(A_i) \cup \text{Select } (k - 1) \text{ highest degree nodes from } V \setminus u$
 - for each node** $v \in \{Solution(A_i) \setminus u\}$ **do**
 - if** $r > 0.5$ **then**
 - $v \leftarrow \text{Replace}(v, N)$.
- 7 $f(Solution(A_i)) \leftarrow$ Calculate the fitness value of each ant $A_i \in A$ ▷ Initialize fitness value of the solution
- 8 Sort all the ant solution $Solution(A_i)$ by their fitness value $f(Solution(A_i))$
- 9 $S_{best} \leftarrow$ Select top $p\%$ solutions based on their fitness value
- 10 Initialize pheromone value of each node $u \in V$ ▷ Initialize pheromone value
- for each solution** $S_i \in S_{best}$ **do**
 - for each node** $u \in S_i$ **do**
 - $\tau_0(u) \leftarrow \tau_0(u) + f(S_i)$
- 11 Evaluate the solution of $\forall A_i \in A$ and update the pheromone value of each node $u \in V$ iteratively ▷ Iteration phase
- while** $(I < I_{max})$ **do**
 - $I \leftarrow I + 1$ ▷ Generate solution for ant
 - for each ant** $A_i \in A$ **do**
 - $Solution(A_i) \leftarrow \phi$
 - $j \leftarrow 1$
 - while** $(j \leq k)$ **do**
 - if** $r < r_0$ **then**
 - $s \leftarrow$ Select node using Equation 9
 - else**
 - $s \leftarrow$ Select node using Equation 10
 - if** $p_{rand} > p_I$ **then**
 - $s \leftarrow$ Randomly select a node $N \in V$
 - $Solution(A_i) \leftarrow Solution(A_i) \cup s$
 - $j \leftarrow j + 1$
 - Evaluate $f(Solution(A_i))$ ▷ Evaluate fitness
 - Sort solutions of all ants based on fitness value ▷ Update pheromone
 - Estimate S_{best}
 - for each solution** $S_i \in S_{best}$ **do**
 - for each node** $u \in S_i$ **do**
 - Update pheromone value of node u using Equation 6
 - 12 $S \leftarrow \text{argmax}_{\forall A_i \in A} \sigma\{Solution(A_i)\}$ ▷ Find best solution
 - 13 **Return** S ▷ Return desired output

Proof Let us assume that all ant solutions contain node u with a new choice of solution, then maximum possible increment in pheromone value $\tau_I(u)$ of node u at any iteration is $\eta(u)$. Therefore, the pheromone $\tau(u)$ at iteration I is bounded due to evaporation and given as

$$\tau(u) \leftarrow (1-\rho)^I \cdot \tau(u) + I_{max} \cdot \sum_{l=1}^{I_{max}} (1-\rho)^{I_{max}-l} \cdot \tau(u), \quad (13)$$

where evaporation rate $0 < \rho \leq 1$; thus, this sum converges to pheromone upper bound $\frac{\tau(u) \cdot I_{max}}{\rho}$ in pheromone update rule. $\square$

Theorem 1 If $\lim_{I \rightarrow \infty} p_I(o_s) = 1$, then the algorithm has reachability and asymptotic convergence, where $p_I(o_s)$ denotes the probability of an algorithm to generate an optimal solution at iteration I .

Proof The minimum pheromone value of any node is greater than 0, because pheromone value is bound with an upper limit as in Proposition 1. The generic solution is evaluated with probability $p_I(o_s) > 0$ at each iteration I since minimum pheromone value is greater than 0. Therefore, after sufficient large number of iterations, the optimal-solution-generating probability tends to 1. Hence, it is guaranteed that the algorithm generates an optimal solution with probability $p_I(o_s)$ close to 1 if enough time is provided to algorithm (asymptotic convergence). $\square$

5 Algorithm

In this section, we give the detailed description of the algorithm ACO-IM. The main Algorithm 1 takes two inputs, an influence graph G and the size of seed set k . In the initialization phase, ants are positioned in solution space with their solution vector based on fitness function and ants start working. In the iteration phase, ants identify the set of feasible solution. Every ant generates the feasible solution and avoids the premature convergence and then evaluates the fitness of solution and updates the pheromone value of nodes. Finally, the best solution is selected as the seed set.

First of all, Lines 1–4 assign values to the all required variables. Line 5 first creates the population size of ants and then randomly distributes these ants into solution space. Line 6 initializes the solution for each ant. In Line 7, ants calculate the fitness value of their solution. Line 8 performs sorting based on solution fitness. Line 9 selects top $p\%$ solutions based on fitness value and stores to S_{best} . Line 10 initializes the pheromone value for each node. Line 11 performs iteration and repeatedly generates solution for each ant. Further, it evaluates fitness of the solution and updates the pheromone value for each node. Line 12 finds the best solution based on influence spread. Finally, the algorithm returns the seed set as the required output.

5.1 Applying the algorithm

In order to explain the working of ACO-IM, we consider an exemplary network with $|V| = 8$ as shown in Fig. 1. Here, we consider ACO-IM system as a complete undirected graph $G(V, E)$, where V represents ants in the colony and E

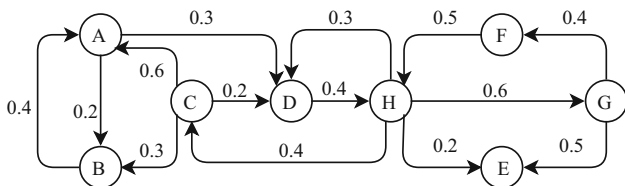


Fig. 1 An exemplary network

represents transmission media of ants to deposit pheromone. Figure 2 illustrates the working of ACO-IM for ant 1. Let us assume that the population of ants is 8, seed size $k = 3$, $\alpha = 0.5$, $\beta = 0.5$, $\rho = 0.2$, and diffusion model $M = LT$.

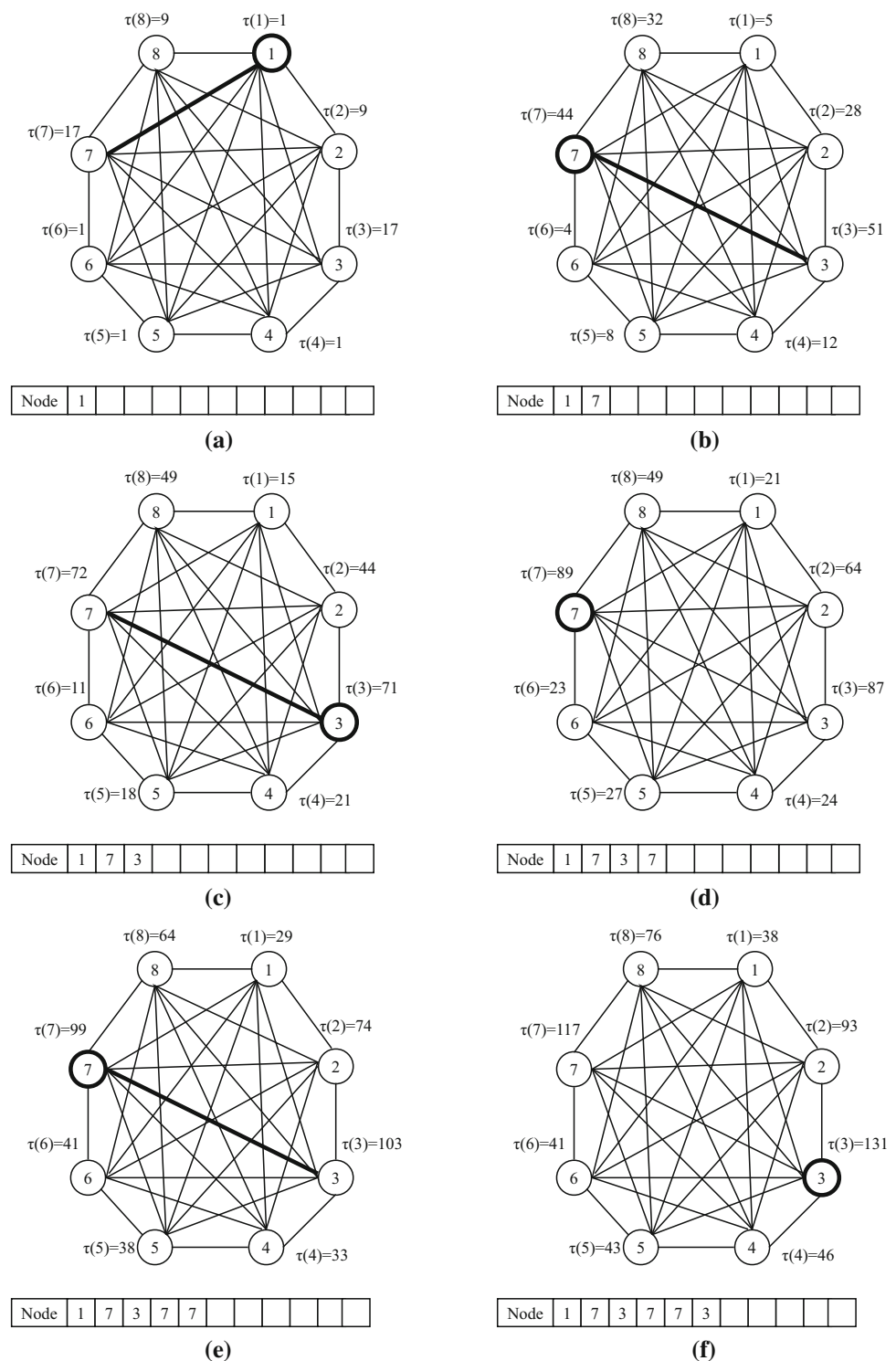
Construct ant solutions Initially, ACO-IM creates solution for each ant based on degree distribution of the nodes. Further, it evaluates the fitness of each solution. Then, the algorithm initializes the pheromone deposited at each node based on top $p\%$ fitness value solutions. For example, ant 1 at node A constructs solution (A, C, G) using Line 6 of the proposed algorithm. Similarly, the algorithm can create the solution of other ants as shown in Table 4. Further, ACO-IM evaluates the fitness of each ant solution and then initializes the pheromone. For example, ant 1 at node A calculates the fitness as of nodes $\{A, C, G\}$ within two-hope area. Figure 3 evaluates the fitness of ant 1 under LT model, i.e., $f(A, C, G) = |A_1^A| = 7$, where active node set of ant A_1 is $A_1^A \leftarrow \{A, C, G, D, B, F, H\}$. Let us assume the fitness values of each ant are given in Table 4. Then, the algorithm initializes the pheromone based on top $p\%$ ($p = 25$), i.e., top 2 solutions as depicted in Table 4. Let Line 2 initialize each node with pheromone value 1. The algorithm updates the pheromone value based on top 2 solutions, i.e., ants 2 and 3. Line 10 updates the pheromone value of nodes B, C, G , and H .

Pheromones update In this phase, algorithm iteratively performs local search and updates pheromone updated by each node. For example, ant 1 positioned at node A decides to move at a new node by applying Eq. 9. Let $r < r_0$ is true, then Eq. 9, i.e., $\text{argmax}_{v \in V \setminus x} \{[\tau(v)]^\alpha \cdot [\eta(v)]^\beta\}$ returns node C . Then, solution of ant 1 is updated, and node C is added to $S(A_1)$. Similarly, ant 1 added G and H to $S(A_1)$ using Eqs. 9 and 10, respectively. Therefore, solution of ant 1 is $S(A_1) \leftarrow \{C, G, H\}$ after iteration 1. The algorithm runs in the same way for each ant in every iteration to estimate optimal solution.

5.2 Time complexity

In this section, we analyze the time complexity of the proposed algorithm ACO-IM. Line 1 takes $O(1)$ basic operation to initialize variable I . Line 2 assigns the pheromone of each node to 1 with $O(|V|)$ basic operations. Line 3 takes $O(1)$ to initialize. Line 4 initializes the solution of each ant to empty set in $O(|R_G|)$ basic operations. Line 5 needs $O(|V|)$ basic operations to create population. Line 6 initializes solution in $O(|R_G| + |V| \log |V|)$ time. Line 7 estimates the fitness of all ants in $O(|R_G| \cdot k \cdot |D_{\text{Avg}}(G)|^2)$ time. Line 8 needs $O(|R_G| \log |R_G|)$ basic operations to perform sorting. Line 9 selects top $p\%$ solution in $O(1)$ time. Line 10 initializes pheromone using $O(|V|)$ basic operations. Line 11 needs $O(I_{\text{max}}(|R_G| \cdot |V| + |R_G| \log |R_G| + |S_{\text{best}}| \cdot |V|))$. There-

Fig. 2 ACO-IM Working: Ant 1 movement over the colony when $|V| = 8$



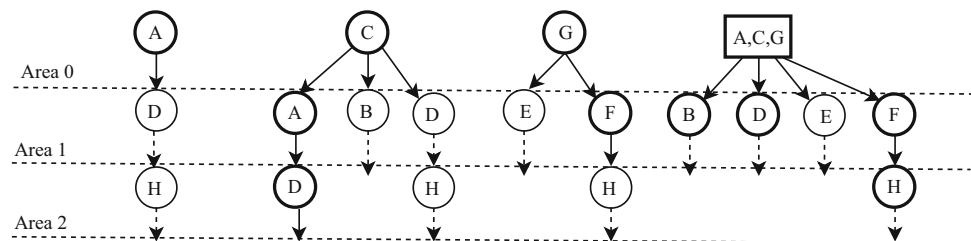
fore, overall complexity of the algorithm is $O(I_{\max}|R_G||V| + |V| \log |V|)$, where $|R_G| \ll |V|$, $|S_{\text{best}}| \ll |R_G|$.

5.3 Scalability and robustness

Most of the IM algorithms are focused on the quality and time efficiency of seed selection procedures, including the proposed algorithm rather than scalability and robustness. Many efforts have been done in the direction of scalability

Table 4 Construct ant solutions: The creation of ant solution with their initial pheromone value

Node N_i	A	B	C	D	E	F	G	H
Node threshold T_i	0.5	0.4	0.5	0.4	0.6	0.3	0.5	0.6
Ant A_i	1	2	3	4	5	6	7	8
Solution $S(A_i)$	A,C,G	B,C,H	C,H,G	D,A,H	E,C,B	F,G,D	G,C,B	H,C,F
Fitness $f(A_i)$	7	8	8	6	5	6	6	6
Pheromone value $\tau(N_i)$	1	9	17	1	1	1	17	9

Fig. 3 Fitness value estimation of ant 1 solution (A,C,G) within two-hope area under LT model

(Li et al. 2019; Nguyen et al. 2019) and robustness (Jung et al. 2012). Li et al. (2019) presented an almost exact solution algorithm for IM on billion-scale networks, named as TipTop. This method generates a $(1 - \epsilon)$ optimal solution on large-scale networks. To improve the efficiency of the method, they reduce the number of samples as much as possible so that the stochastic programming can be solved in a short time. Nguyen et al. (2019) proposed a CPU-GPU method to improve the scalability on large networks, viz. hitting self-avoiding walks (HSAWs). They adopt traditional LT model to propagate rumors. They propose three acceleration techniques for distributed and parallel algorithms on GPUs.

Jung et al. (2012) proposed a more general PageRank method named as IRIE. This method gives some ranking to each user based on message passing influence estimation. It designs a system of $|V|$ equations with $|V|$ varying to compute the influence of each $x \in V$. To estimate influence spread of x , it combines the influence of itself with its direct influence to neighborism, i.e., $(1 + \alpha \sum_{y \in N_{out}(x)} p_{xy} \cdot Inf_y)$, where $\alpha \in [0, 1]$. This method solves the system of linear equation iteratively, and after k iterations it returns the seed set S . IRIE is highly robust and matches the influence spread with greedy with high efficiency.

Table 5 compares the performance metrics of the proposed method against the state-of-the-art methods. Quality, in IM problem, equates the number of product adoption in network by algorithm, with given seed set S , $|S| = k$. Efficiency, in IM problem, is measured in terms of running time. Efficiency is the ability of an algorithm to produce a desired result, i.e., seed set S in efficient time. Robustness is equally an important aspect of performance measure of an IM algorithm. An algorithm is called as robust if the optimal seed set does not change much when a slight change occurs in diffusion model. Scalability of an IM algorithm is measured in terms of both running time and memory consumption.

6 Experiment

In this section, firstly, the experimental setup information is provided along with the dataset and probability distribution for diffusion models. Then, we discuss the compared algorithms. Further, we discuss the performance analysis regarding influence spread and efficiency along with parameter analysis. Furthermore, we illustrate the statistical test to show the significant difference between the proposed method and the compared methods. The classical methods, e.g., CELF++, utilize the MC simulations to measure the performance of algorithms.

6.1 Setup

We compare the proposed algorithm with six algorithms: one greedy algorithm (CELF++), two heuristics (Degree Discount and Diffusion Degree), and three swarm-based meta-heuristics (PSO, IM-SSO, and LAPSO-IM). We use six real-world social networks to perform experiments. To assign influence probabilities, we utilize well-known probability distributions. The propagation probability for each edge follows a uniform distribution. The propagation probability in IC models is assigned based on uniform distribution over $\{0.1, 0.01, 0.001\}$. For the remaining two models (WC and LT), propagation probability $p_{x,y}$ is assigned by $\frac{1}{\text{indegree}(y)}$. The activation probability of a node x follows uniform distribution over $[0, 1]$. Each of the methods is implemented using C++ language and executed on DEV-C++. All of the algorithms run 20 times independently on a 64-bit Window's PC with Intel(R) Core(TM) i7-7700 CPU@ 3.60GHz processor and 8GB memory.

Table 5 Comparison of the performance metrics of the state-of-the-art algorithms with the proposed algorithm

Algorithm	Time efficiency	Seed quality	Memory footprint	Robustness	Scalability
Greedy (Kempe et al. 2003)	✗	✓	✓	✗	✗
Knapsack Greedy (Sviridenko 2004)	✗	✓	✓	✗	✗
SP1M (Kimura and Saito 2006)	✓	✗	✓	✗	✓
CELF (Leskovec et al. 2007)	✗	✓	✓	✗	✗
Degree discount (Chen et al. 2009)	✓	✓	✓	✗	✓
TW Greedy (Wang and Feng 2009)	✗	✓	✓	✗	✗
CGA (Wang et al. 2010)	✓	✓	✓	✗	✓
CELF++ (Goyal et al. 2011)	✗	✓	✓	✗	✗
Diffusion degree (Kundu et al. 2011)	✓	✓	✓	✗	✓
SIMPATH (Goyal et al. 2011)	✓	✓	✓	✗	✓
IRIE (Jung et al. 2012)	✗	✗	✓	✓	✓
IPA (Kim et al. 2013)	✓	✗	✓	✗	✓
StaticGreedy (Cheng et al. 2013)	✗	✓	✗	✗	✓
PRUNEDMC (Ohsaka et al. 2014)	✗	✓	✗	✗	✓
TIM (Tang et al. 2014)	✗	✓	✗	✗	✓
PSO (Gong et al. 2016)	✓	✓	✓	✗	✓
LAIM (Ge et al. 2017)	✗	✓	✓	✗	✓
IM-SSO (Singh et al. 2019a)	✓	✓	✓	✗	✓
PSO (Singh et al. 2019b)	✓	✓	✓	✗	✓
TipTop (Li et al. 2019)	✓	✓	✗	✗	✓[High]
CPU-GPU-HSAW (Nguyen et al. 2019)	✓	✓	✗	✗	✓[High]
ACO-IM	✓	✓	✓	✗	✓

6.2 Dataset

NetInfective¹ (Isella et al. 2011) network describes people's behavior during face-to-face interaction at INFECTIOUS exhibition. Gnutella30² (Ripeanu et al. 2002) is a peer-to-peer file sharing network. The other networks NetScience¹ (Newman 2006), Ca-GrQc³ (Leskovec et al. 2007), NetHEPT¹ (Chen et al. 2009), and NetPHY¹ (Chen et al. 2013) are co-author networks and describe authors relationship among various research domains. Table 6 describes the statistical information of the real-world datasets.

6.2.1 Seeding strategies to compare

The description of the compared seeding algorithms is given as follows:

- **Degree discount**(Chen et al. 2009) The algorithm iteratively selects the seed based on the degree of the node. After selecting each seed algorithm performs degree discount step which reduces the degree of newly activated

node's neighbors by one. This is because newly active node is no longer activated by others.

- **Diffusion degree**(Kundu et al. 2011). The algorithm selects the seed nodes based on their local and global influence. The local influence of a node is evaluated based on influence to direct neighbors, while global influence focuses on the indirect influence, i.e., neighbors to rest of the network.
- **CELF++** (Goyal et al. 2011) The algorithm is the extension of CELF and utilizes the objective function's sub-modular property to reduce the search space. This algorithm is 30–50% faster than CELF.
- **PSO** (Gong et al. 2016) This meta-heuristic uses discrete particle swarm optimization to approximate the seed users.
- **IM-SSO** (Singh et al. 2019a) This meta-heuristic uses social spider optimization to select seed users.
- **LAPSO-IM** (Singh et al. 2019b) This meta-heuristic uses discrete particle swarm optimization to identify influential users. A learning strategy is also utilized to avoid premature convergence.
- **ACO-IM** This is our proposed algorithm. It uses ant colony optimization to approximate the solution.

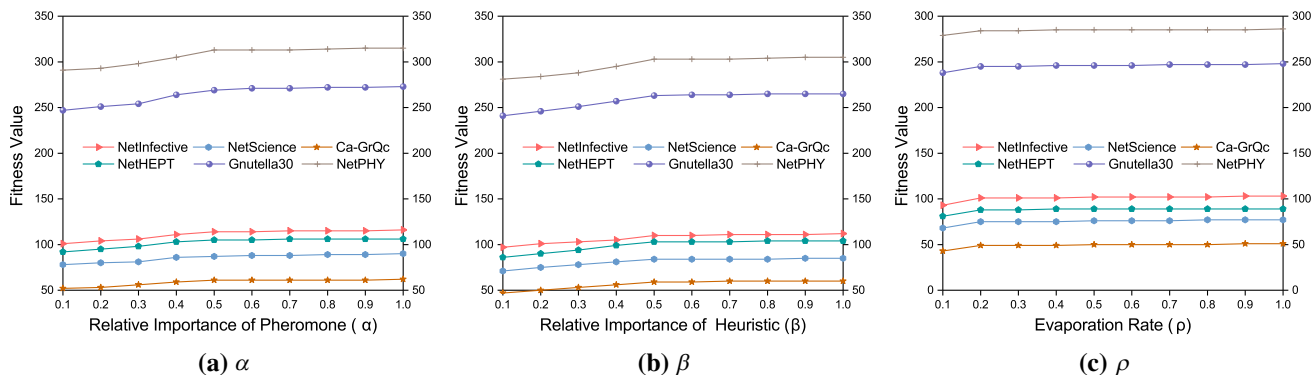
¹ <https://arxiv.org/>.

² <https://snap.stanford.edu/data/p2p-Gnutella30.html>.

³ <http://snap.stanford.edu/data/ca-GrQc.html>.

Table 6 Statistical information of the real-world datasets

Dataset	$ V $	$ E $	$D_{\text{avg}}(N)$	Type
NetInfective (Isella et al. 2011)	410	17298	84.38	Dense
NetScience (Newman 2006)	1589	2742	3.45	Sparse
Ca-GrQc (Leskovec et al. 2007)	5242	14496	5.53	Sparse
NetHEPT (Chen et al. 2009)	15233	31398	4.12	Sparse
Gnutella30 (Ripeanu et al. 2002)	36682	88328	4.81	Sparse
NetPHY (Chen et al. 2013)	37154	174161	9.38	Sparse

**Fig. 4** Variation of fitness value regarding various parameters

6.3 Parameter setting

The parameters ρ , α , and β have been taken in the proposed algorithm. The parameters α and β are related to the importance of pheromone and quality of node. We tested both α and β , ranging from 0 to 1, and found that the fitness value increases with the increase in the value of both the parameters. To assign the parameter value of α , we set the value of other parameters $\beta = 0.5$, $\rho = 0.2$. With this, we perform experiments on different datasets and found that the fitness value increases with the increasing value of $\alpha = 0.1 - 0.5$, but fitness value is approximate with $\alpha > 0.5$ as shown in Fig. 4a. Similarly, we test the parameter value of β with $\alpha = 0.5$, $\rho = 0.2$ setting as shown in Fig. 4b. Therefore, ACO-IM gives optimal result at $\alpha = 0.5$, $\beta = 0.5$. The evaporation rate parameter ρ is assigned to 0.2. This is because it allows the pheromone value to evaporate too quickly. Therefore, ACO-IM searches all possible nodes. To assign the parameter value of ρ , we set the $\alpha = 0.5$ and $\beta = 0.5$ as shown in Fig. 4c.

6.4 Performance analysis

6.4.1 Quality

In order to measure the performance of the proposed algorithm regarding quality or effectiveness, we equate influence spread on six real-world social networks for different size seed set (10–50). Figures 5, 6, and 7 demonstrate the

comparison of influence spread of the proposed algorithm under classical diffusion models (IC, WC, and LT) against best-suited compared seeding algorithms. ACO-IM produces better-quality seed nodes compared to heuristic-based algorithms Degree Discount and Diffusion Degree. This is because ACO-IM utilizes more information such as degree, local influence evaluation metrics, and local search procedure compared to both heuristics. The ACO-IM compromises very little in quality to improve efficiency compared to CELF++. Therefore, influence spread of the proposed method is slightly lower than greedy-based method CELF++. The influence spread of ACO-IM is better than both PSO and IM-SSO and comparable with LAPSO-IM. This is because LAPSO-IM utilizes the learning automata to improve the search strategy and avoids local optima.

In Table 7, we summarize the ANP (the percentage of nodes activated in network) and NPS (the average number of nodes activated by each seed in network), respectively. For example, the proposed algorithm activates about 32.44% nodes in the NetInfective network and each seed activates about 2.66 nodes under IC diffusion model. Table 8 gives the information of speedup % of the proposed algorithm over the state-of-the-art algorithm. For example, ACO-IM has up to 48% more influence spread than Degree Discount under IC model, i.e., 1.4 times better than Degree Discount.

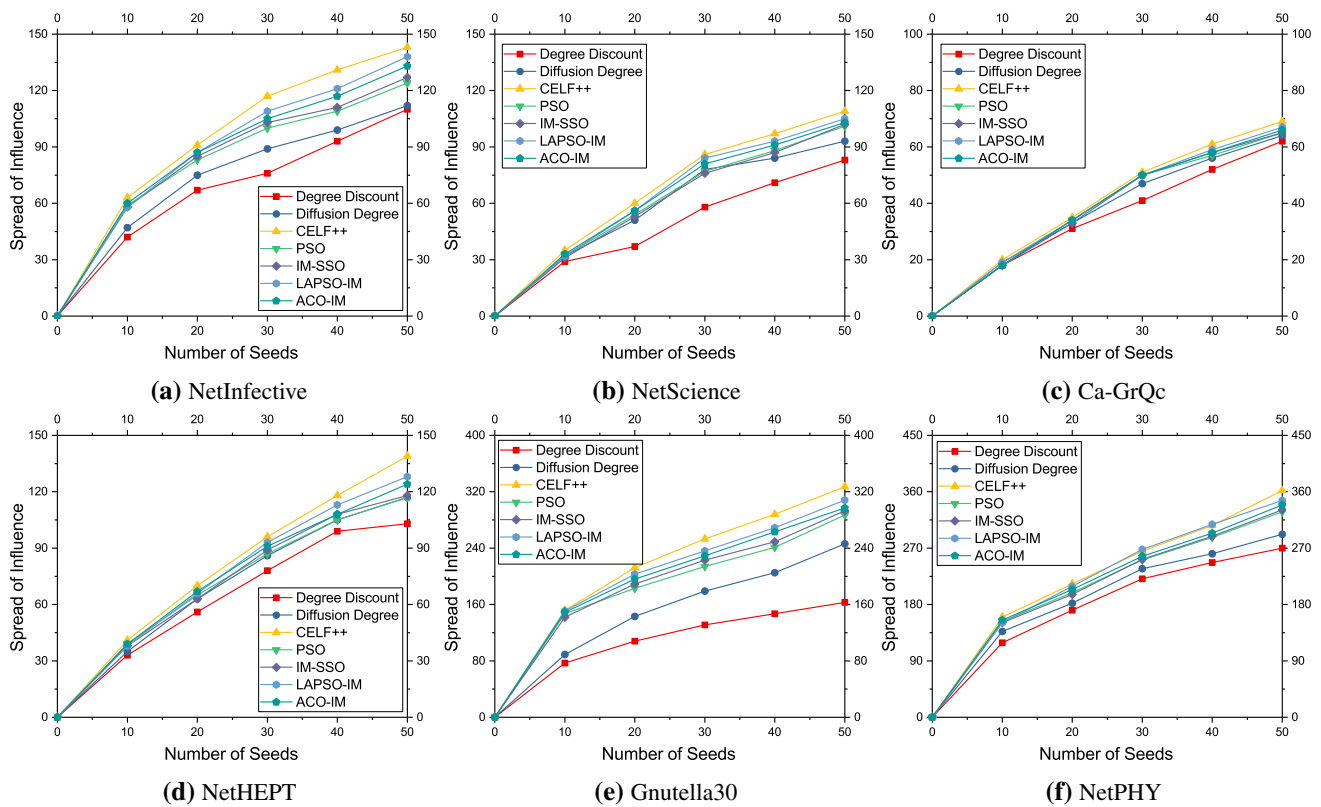


Fig. 5 Influence spread comparison in various datasets under IC model

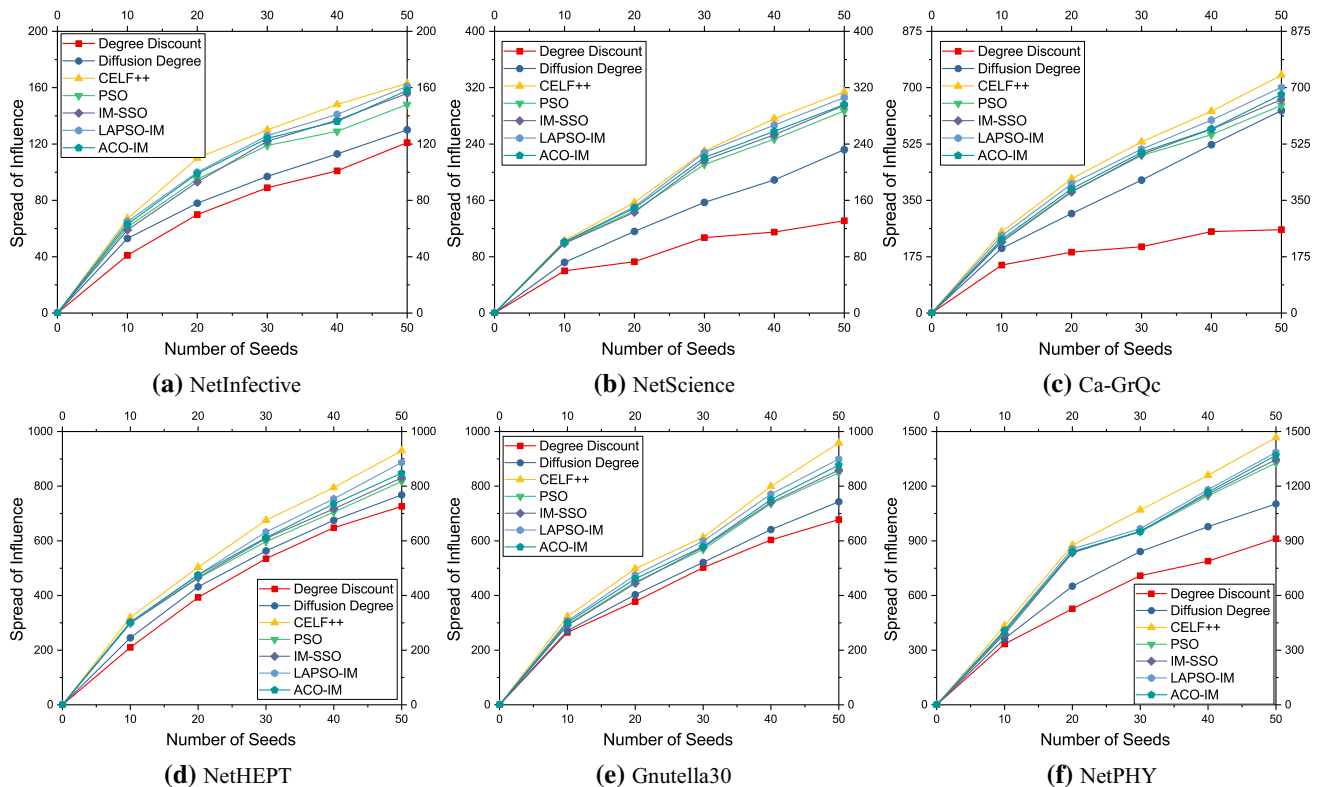


Fig. 6 Influence spread comparison in various datasets under WC model

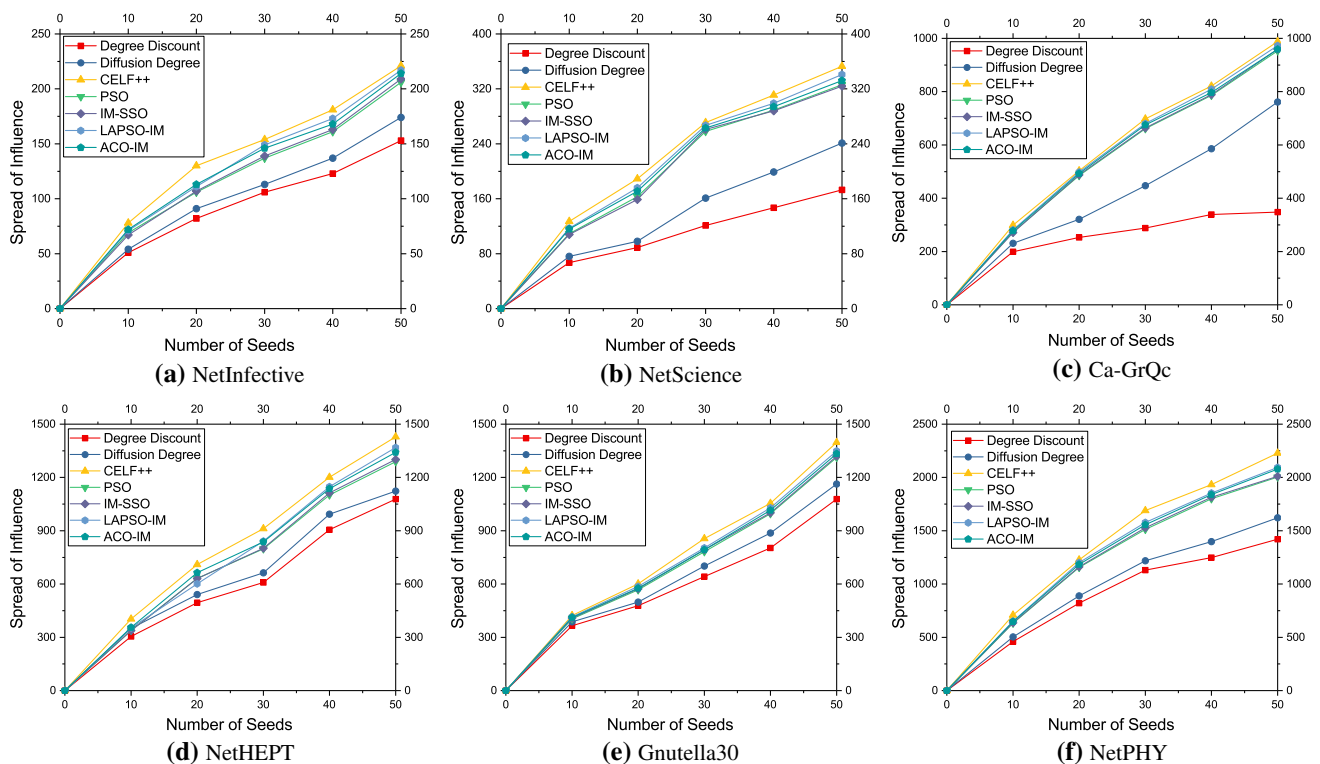


Fig. 7 Influence spread comparison in various datasets under LT model

6.4.2 Efficiency

To compare the efficiency of the proposed algorithm against compared seeding strategies, we utilize the same datasets used in performance analysis regarding quality. We perform experiments under classical diffusion models with seed set $|S| = 50$ in six real-world social networks. Figure 8 illustrates the efficiency comparison of the ACO-IM against the compared methods under classical diffusion models. Heuristic-based methods (Degree Discount and Diffusion Degree) are much faster than ACO-IM because both methods only utilize topological information like degree and avoid any searching strategy. Therefore, both methods are more efficient than ACO-IM with poor-quality seed set. The proposed method has almost the same efficiency as meta-heuristic methods (PSO, IM-SSO, and LAPSO-IM) with comparable influence spread. ACO-IM outperforms CELF++ regarding efficiency because CELF++ performs time-consuming MC simulations. Therefore, ACO-IM achieves a trade-off between influence spread and running time as shown in Figs. 5, 6, 7, and 8.

6.5 Statistical test

The statistical tests are performed in this section to show the significant difference between ACO-IM and the state-of-the-art methods. First, Friedman test (Friedman 1937; Derrac et al. 2011) on average influence spread is performed to verify the significant difference. After we find a significant difference between algorithm, Holm–Bonferroni procedure (Derrac et al. 2011) is applied as a post hoc procedure. The post hoc procedure estimates the p value as the degree of rejection of each hypothesis. The proposed algorithm ACO-IM is considered as a control algorithm in the post hoc test with a degree of freedom $D_f = 6$ and level of confidence $\alpha_c = 0.05$ in all the cases.

To perform the Friedman test, we need to state the null hypothesis (H_0). The H_0 states that all the compared algorithms are statistically equivalent, i.e., there is no significant difference between them. The null hypothesis is rejected only if $F_f > \chi^2(\alpha_c, D_f)$, i.e., $F_f > 12.592$, where F_f denotes the test statistic value. Table 9 demonstrates that the null hypothesis is rejected based on the Friedman test under IC model for distinct seed sets. Therefore, there is a significant

Table 7 Comparison of ANP (the percentage of activated nodes in original social network) and NPS (the average number of nodes activated per seed) under traditional diffusion models (IC, WC, and LT) in various datasets for $|S| = 50$

Model	Dataset	Degree discount		Diffusion degree		CELF++		PSO		IM-SSO		LAPSO-IM		ACO-IM	
		ANP	NPS	ANP	NPS	ANP	NPS	ANP	NPS	ANP	NPS	ANP	NPS	ANP	NPS
IC	NetInfective	26.83	2.20	27.32	2.24	34.88	2.86	30.24	2.48	30.98	2.54	33.66	2.76	32.44	2.66
	NetScience	5.22	1.66	5.85	1.86	6.86	2.18	6.36	2.02	6.42	2.04	6.61	2.10	6.48	2.06
	GrQc	1.18	1.24	1.22	1.28	1.32	1.38	1.24	1.30	1.24	1.30	1.28	1.34	1.26	1.32
	NetHEPT	0.68	2.06	0.77	2.34	0.91	2.78	0.77	2.34	0.77	2.36	0.84	2.56	0.81	2.48
	Gnutella30	0.44	3.26	0.67	4.92	0.89	6.54	0.78	5.74	0.80	5.86	0.84	6.16	0.81	5.94
WC	NetPHY	0.73	5.40	0.79	5.84	0.97	7.24	0.88	6.56	0.89	6.62	0.93	6.92	0.91	6.78
	NetInfective	29.51	2.42	31.71	2.60	39.76	3.26	36.10	2.96	38.05	3.12	39.27	3.22	38.54	3.16
	NetScience	8.24	2.62	14.60	4.64	19.76	6.28	18.06	5.74	18.57	5.90	19.26	6.12	18.63	5.92
	GrQc	4.94	5.18	12.00	12.58	14.10	14.78	12.30	12.90	12.65	13.26	13.37	14.02	12.95	13.58
	NetHEPT	4.77	14.52	5.04	15.36	6.11	18.60	5.36	16.34	5.46	16.62	5.82	17.74	5.57	16.96
LT	Gnutella30	1.85	13.56	2.03	14.86	2.61	19.16	2.31	16.98	2.34	17.18	2.45	17.98	2.39	17.54
	NetPHY	2.45	18.24	2.97	22.06	3.95	29.34	3.57	26.56	3.62	26.92	3.73	27.72	3.68	27.38
	NetInfective	37.32	3.06	42.44	3.48	53.90	4.42	50.24	4.12	50.98	4.18	52.93	4.34	52.20	4.28
	NetScience	10.89	3.46	15.17	4.82	22.22	7.06	20.52	6.52	20.39	6.48	21.46	6.82	20.89	6.64
	GrQc	6.64	6.96	14.52	15.22	18.85	19.76	18.18	19.06	18.29	19.18	18.56	19.46	18.29	19.18
	NetHEPT	7.08	21.56	7.37	22.46	9.38	28.58	8.46	25.78	8.54	26.02	8.98	27.36	8.80	26.82
	Gnutella30	2.94	21.56	3.17	23.26	3.81	27.94	3.58	26.24	3.60	26.38	3.68	26.98	3.64	26.68
	NetPHY	3.82	28.42	4.36	32.42	6.00	44.56	5.39	40.06	5.41	40.22	5.64	41.88	5.59	41.56

Table 8 Influence spread speedup % of ACO-IM over state-of-the-art algorithms in different datasets

Diffusion Model	Algorithm Compared	Seed Set size	Datasets					
			NetInfective	NetScience	Ca-GrQc	NetHEPT	Gnutella30	NetPHY
IC	Degree discount	10	30.00	12.12	0.00	15.38	48.32	23.23
		20	22.99	33.93	8.82	16.42	44.90	16.18
		30	27.62	28.40	18.00	14.29	42.79	14.01
		40	20.51	21.98	10.34	8.33	44.11	15.99
		50	17.29	19.42	6.06	16.94	45.12	20.35
	Diffusion degree	10	21.67	3.03	− 5.56	10.26	40.27	11.61
		20	13.79	8.93	2.94	5.97	27.04	10.78
		30	15.24	3.70	6.00	5.49	21.83	7.78
		40	15.38	7.69	3.45	2.78	22.05	11.22
		50	15.79	9.71	3.03	5.65	17.17	13.86
	CELF++	10	− 5.00	− 6.06	− 11.11	− 5.13	− 2.01	− 3.23
		20	− 4.60	− 7.14	− 2.94	− 4.48	− 8.16	− 4.41
		30	− 11.43	− 6.17	− 2.00	− 5.49	− 10.48	− 3.11
		40	− 11.97	− 6.59	− 5.17	− 9.26	− 9.51	− 4.08
		50	− 7.52	− 5.83	− 4.55	− 12.10	− 10.10	− 6.78
	PSO	10	1.67	3.03	0.00	0.00	1.34	1.29
		20	4.60	3.57	0.00	2.99	6.63	2.45
		30	4.76	4.94	0.00	4.40	6.55	2.33
		40	6.84	3.30	1.72	2.78	8.37	2.38
		50	6.77	1.94	1.52	5.65	3.37	3.24
	IM-SSO	10	3.33	6.06	0.00	2.56	4.70	2.58
		20	2.30	5.36	2.94	5.97	3.57	3.92
		30	1.90	6.17	0.00	2.20	2.62	1.95
		40	5.13	4.40	0.00	0.00	5.32	1.70
		50	4.51	0.97	1.52	4.84	1.35	2.36
	LAPSO-IM	10	3.33	3.03	− 5.56	2.56	− 1.34	2.58
		20	0.00	0.00	0.00	1.49	− 3.57	− 2.45
		30	− 3.81	− 3.70	0.00	− 2.20	− 3.06	− 4.28
		40	− 3.42	− 2.20	− 1.72	− 4.63	− 2.28	− 4.76
		50	− 3.76	− 1.94	− 1.52	− 3.23	− 3.70	− 2.06
WC	Degree discount	10	34.92	40.59	34.93	30.69	11.96	17.73
		20	29.29	51.01	51.04	17.26	18.00	37.34
		30	28.23	51.58	58.72	12.58	13.30	25.45
		40	25.74	55.43	55.77	11.84	19.71	32.39
		50	23.42	55.74	61.86	14.39	22.69	33.38
	Diffusion degree	10	15.87	28.71	12.23	19.14	9.97	10.59
		20	21.21	22.15	19.95	9.05	12.58	22.59
		30	21.77	28.96	17.23	8.01	10.02	11.57
		40	16.91	26.74	8.57	8.16	14.65	16.20
		50	17.72	21.62	7.36	9.43	15.28	19.43
	CELF++	10	− 6.35	− 1.98	− 10.04	− 5.61	− 7.31	− 6.90
		20	− 11.11	− 5.37	− 8.03	− 5.89	− 8.03	− 4.16
		30	− 4.84	− 4.07	− 6.61	− 10.29	− 5.87	− 12.51
		40	− 8.82	− 6.98	− 9.62	− 8.16	− 6.39	− 7.97
		50	− 3.16	− 6.08	− 8.84	− 9.67	− 9.24	− 7.16

Table 8 continued

Diffusion Model	Algorithm Compared	Seed Set size	Datasets					
			NetInfective	NetScience	Ca-GrQc	NetHEPT	Gnutella30	NetPHY
LT	PSO	10	3.17	0.99	2.62	1.32	2.33	1.97
		20	4.04	2.68	2.07	2.53	2.60	0.36
		30	4.03	4.52	2.00	2.45	1.90	0.21
		40	5.15	4.26	3.15	4.08	2.00	1.71
		50	6.33	3.04	5.01	3.66	3.19	2.99
	IM-SSO	10	6.35	1.98	3.49	1.65	3.32	4.19
		20	6.06	4.03	2.59	1.89	3.69	0.83
		30	1.61	1.81	1.20	0.49	0.52	− 0.21
		40	− 0.74	1.94	0.17	2.45	1.33	0.86
		50	1.27	0.34	2.36	2.00	2.05	1.68
	LAPSO-IM	10	− 3.17	0.00	− 4.80	1.32	− 1.99	− 1.23
		20	− 1.01	− 1.34	− 3.89	− 0.21	− 3.04	− 1.78
		30	− 1.61	− 3.17	− 2.00	− 3.27	− 3.45	− 1.68
		40	− 3.68	− 3.49	− 4.72	− 2.59	− 2.66	− 1.20
		50	− 1.90	− 3.38	− 3.24	− 4.60	− 2.51	− 1.24
	Degree discount	10	29.17	42.24	28.42	14.33	11.41	29.01
		20	27.43	47.95	48.68	25.45	17.01	30.74
		30	27.40	53.99	57.33	27.24	19.04	27.38
		40	26.79	50.00	57.47	20.32	20.89	32.14
		50	28.50	47.89	63.71	19.61	19.19	31.62
	Diffusion degree	10	25.00	34.48	16.91	1.40	6.07	22.38
		20	19.47	42.69	34.89	18.52	13.54	24.92
		30	22.60	38.78	33.78	20.79	11.60	21.72
		40	18.45	32.31	26.47	12.66	12.61	23.98
		50	18.69	27.41	20.65	16.26	12.82	21.99
	CELF++	10	− 8.33	− 9.48	− 7.55	− 13.20	− 2.67	− 9.41
		20	− 15.04	− 10.53	− 2.03	− 6.93	− 4.34	− 3.55
		30	− 5.48	− 3.04	− 3.41	− 8.96	− 7.94	− 8.55
		40	− 7.74	− 5.78	− 3.01	− 5.63	− 3.84	− 5.06
		50	− 3.27	− 6.33	− 3.02	− 6.56	− 4.72	− 7.22
	PSO	10	4.17	6.03	1.80	4.78	1.94	2.31
		20	6.19	4.68	1.62	4.82	1.56	2.36
		30	6.16	1.90	2.07	4.66	1.39	2.76
		40	4.17	1.70	1.38	3.17	1.87	2.18
		50	3.74	1.81	0.63	3.88	1.65	3.61
	IM-SSO	10	6.94	6.90	2.52	5.06	0.97	1.70
		20	5.31	7.02	1.01	4.97	0.52	1.94
		30	4.79	0.76	1.63	4.30	0.00	1.80
		40	2.98	2.04	1.00	2.11	1.38	1.36
		50	2.34	2.41	0.00	2.98	1.12	3.22
	LAPSO-IM	10	1.39	− 0.86	− 1.08	0.84	− 0.73	− 0.46
		20	1.77	− 2.92	− 1.01	9.49	− 1.91	− 1.44
		30	− 2.05	− 1.52	− 0.74	− 0.48	− 1.26	− 1.41
		40	− 2.98	− 1.70	− 1.51	− 0.97	− 1.38	− 0.82
		50	− 1.40	− 2.71	− 1.46	− 2.01	− 1.12	− 0.77

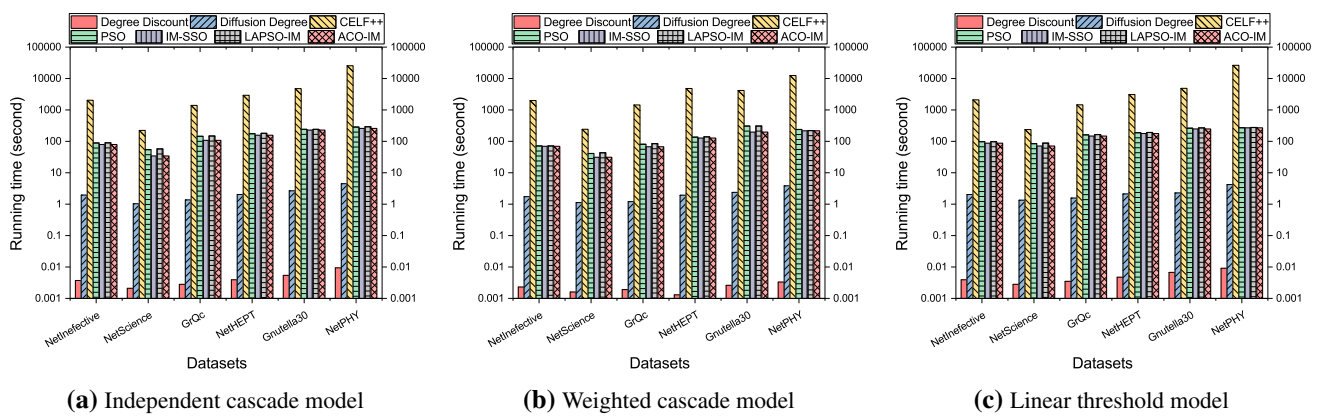


Fig. 8 Running time comparison in various datasets under traditional diffusion models for $|S| = 50$

Table 9 Friedman test on average influence spread under IC diffusion model

Seed	Dataset	IS value							Test value F_f	State result Is $F_f > \chi^2$?
		Degree discount	Diffusion degree	CELF++	PSO	IM-SSO	LAPSO-IM	ACO-IM		
10	NetInfective	42	47	63	59	58	58	60	27.691	Null Hypothesis Rejected
	NetScience	29	32	35	32	31	32	33		
	GrQc	18	19	20	18	18	19	18		
	NetHEPT	33	35	41	39	38	38	39		
	Gnutella30	77	89	152	147	142	151	149		
	NetPHY	119	137	160	153	151	151	155		
20	NetInfective	67	75	91	83	85	87	87	34.39	Null Hypothesis Rejected
	NetScience	37	51	60	54	53	56	56		
	GrQc	31	33	35	34	33	34	34		
	NetHEPT	56	63	70	65	63	66	67		
	Gnutella30	108	143	212	183	189	203	196		
	NetPHY	171	182	213	199	196	209	204		
30	NetInfective	76	89	117	100	103	109	105	33.055	Null Hypothesis Rejected
	NetScience	58	78	86	77	76	84	81		
	GrQc	41	47	51	50	50	50	50		
	NetHEPT	78	86	96	87	89	93	91		
	Gnutella30	131	179	253	214	223	236	229		
	NetPHY	221	237	265	251	252	268	257		
40	NetInfective	93	99	131	109	111	121	117	34.937	Null Hypothesis Rejected
	NetScience	71	84	97	88	87	93	91		
	GrQc	52	56	61	57	58	59	58		
	NetHEPT	99	105	118	105	108	113	108		
	Gnutella30	147	205	288	241	249	269	263		
	NetPHY	247	261	306	287	289	308	294		
50	NetInfective	110	112	143	124	127	138	133	35.802	Null Hypothesis Rejected
	NetScience	83	93	109	101	102	105	103		
	GrQc	62	64	69	65	65	67	66		
	NetHEPT	103	117	139	117	118	128	124		
	Gnutella30	163	246	327	287	293	308	297		
	NetPHY	270	292	362	328	331	346	339		

Table 10 Estimation of p value (unadjusted) based on the Holm procedure for post hoc analysis on average influence spread under traditional diffusion models

Model	Seed	p value					
		Degree discount	Diffusion degree	CELF++	PSO	IM-SSO	LAPSO-IM
IC	10	1.50E-06	0.00175	0.00888	0.21308	0.00245	0.21308
	20	< 2E-16	< 2E-16	1.20E-09	1.10E-08	2.70E-12	0.3933
	30	1.90E-12	2.10E-08	4.40E-06	3.40E-05	0.0019	0.00661
	40	< 2E-16	< 2E-16	< 2E-16	< 2E-16	3.60E-14	< 2E-16
	50	< 2E-16	< 2E-16	< 2E-16	2.07E-13	1.94E-14	< 2E-16
WC	10	1.70E-06	0.00757	0.07033	1.0832	0.00342	0.08851
	20	1.8E-13	< 2E-16	1.20E-09	1.35E-08	2.20E-12	0.27872
	30	1.70E-12	1.80E-08	3.60E-06	3.40E-05	0.06408	0.00548
	40	< 2E-16	< 2E-16	< 2E-16	< 2E-16	3.60E-14	< 2E-16
	50	< 2E-16	< 2E-16	< 2E-16	2.5E-13	1.70E-14	< 2E-16
LT	10	2.20E-05	0.00548	0.02006	0.92580	0.00435	0.09022
	20	1.8E-13	< 2E-16	1.40E-09	1.6E-08	2.60E-12	0.25455
	30	1.70E-12	1.80E-08	3.60E-06	3.40E-05	0.06408	0.00548
	40	< 2E-16	< 2E-16	< 2E-16	< 2E-16	3.60E-14	< 2E-16
	50	< 2E-16	< 2E-16	< 2E-16	2E-14	1.90E-14	< 2E-16

difference between ACO-IM and baseline methods on average influence spread. Similarly, we applied the Friedman test under remaining diffusion (WC and LT) models for different seed sets and H_0 is rejected in each case. Now, we perform the Holm–Bonferroni test to measure the level of significant difference.

Table 10 illustrates the unadjusted p values under classical diffusion models for average influence spread values. These p values show the level of significant differences between the control method ACO-IM and compared seeding methods. The test values state that ACO-IM is approximate to CELF++ and LAPSO-IM algorithms and outperforms the remaining four algorithms. However, the control method is comparable to all methods except CELF++ regarding running time. Therefore, ACO-IM achieves a trade-off between quality and efficiency.

7 Conclusion and future directions

In this paper, we have proposed an influence maximization algorithm based on ant colony optimization to maximize influence spread. First, we have presented a local influence heuristic by extending *expected diffusion value* function to within the two-hop area. Further, we adopted ant colony optimization by redefining the representation and update rule of pheromone deposited by ants and heuristic information for seed selection. To find the new position of ant, the proposed algorithm used a probabilistic process. The probabilistic process avoids ACO-IM getting stuck in local optima. Finally,

the experimental evaluation is presented to analyze the performance of the proposed work.

The experimental results on six real-world networks demonstrated that the ACO-IM performed better than both heuristic methods (Degree Discount & Diffusion Degree) and meta-heuristic (PSO & IM-SSO) regarding influence spread with comparable efficiency. ACO-IM performed up to 1.48, 1.55, and 1.63 times better than heuristic methods w.r.t.⁴ influence spread under IC, WC, and LT diffusion models, respectively. LAPSO-IM has approximate influence spread with comparable efficiency. The proposed algorithm performed up to 1.08, 1.06, and 1.06 times better than meta-heuristic methods w.r.t influence spread under IC, WC, and LT diffusion models, respectively. ACO-IM is more time-efficient than CELF++ with approximate influence spread and comparable efficiency with remaining algorithms. Therefore, the proposed algorithm is a trade-off between influence spread and time efficiency. The statistical test illustrates that the proposed algorithm has a significant difference from existing algorithms.

Much future technical work is possible; for example, the proposed algorithm can be extended into the multi-objective function, e.g., location, budget, time, competitiveness, topic, target, etc., to achieve a better trade-off between accuracy and efficiency. Further, we will adopt learning automata-induced swarm optimization techniques to incorporate noisy fitness. Furthermore, we will focus on multiple influence maximization and numerous social networks along with heterogeneous

⁴ With respect to

diffusion models to include more realistic and practical scenario.

Compliance with ethical standards

Conflict of interest The authors declare no conflict of interest.

Human or animal rights The article does not contain any studies with human or animal subjects. This article presents an algorithm ACO-IM.

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